

Robyn Jorgensen, Shelley Dole, Kevin Larkin

TEACHING MATHEMATICS IN PRIMARY SCHOOLS

PRINCIPLES FOR EFFECTIVE PRACTICE

Teaching Mathematics in Primary Schools



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Principles for effective practice

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Robyn Jorgensen, Shelley Dole and Kevin Larkin

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ABOUT THIS BOOK

This book has been designed to introduce pre-service and practising teachers to current methods of teaching school mathematics to primary and middle-years students. We have drawn on a wide range of research to inform the book, but have not taken one specific approach. Rather, our intention has been to expose readers to the diversity of approaches to teaching school mathematics. Our rationale for doing this is that students learn from a range of experiences, and, with such diversity in contemporary classrooms, teachers need to have a wide repertoire of skills and knowledge if they are to provide the quality experiences needed for student success. It is recognised that the most important factor in creating success for learners is the teacher. As such, teachers in today's society have a critical role in facilitating successes for all students. By having a rich repertoire and a strong knowledge based on evidence, teachers are better able to prepare their students for a positive future.

We have opted to use the term 'school mathematics' rather than 'mathematics' because there is increasing recognition that what is taught in schools is a particular form of mathematics. It is quite different from the mathematics used in workplaces or by mathematicians, and should be recognised as such. Creating opportunities for students to learn and develop a deep appreciation for mathematical thinking and doing is essential for success in life. School mathematics plays an important role in the preparation of young people for their entry into the adult world.

Trends in research are reflected by the availability or paucity of relevant publications. In this third edition, we have updated the book with more current literature—particularly through an increased emphasis on the use of digital technologies in mathematics education. In this edition, we have also consolidated various aspects of numeracy found throughout the second edition into a dedicated chapter on numeracy (Chapter 8), in which we examine numeracy from a range of perspectives and distinguish between mathematics and numeracy.

We spent some time considering the merits of including a stand-alone chapter on the use of digital technologies in the teaching of school mathematics; however, we have opted instead to incorporate this topic into each chapter. This is so that readers will be able to see where and how technologies (digital and otherwise) can be used in the teaching of school mathematics. We argue strongly that, when used appropriately, technologies are useful tools in the enhancement of mathematical learning (Larkin 2016). They enable deeper explorations of many ideas by removing the tedium of calculations or drawings, and can extend children's mathematical understanding, particularly in geometry. However, technologies can also be used inappropriately (e.g. as a form of reward or as a time filler). We address these issues within each chapter when we propose various digital technologies to support learning and teaching.

A continued theme in this third edition is the inclusion of the 'middle years'—those years between upper primary and early secondary—since many schools maintain specific programs for learners in these years. For many students, the transition to secondary school involves revising much of the mathematics taught in primary school, creating the risk of significant repetition in learning, rather than being seen as part of the learning continuum. To help avoid this considerable overlap, we have included the extra school mathematics that is typically taught in these years, and provided new links to digital resources that can support this new mathematical content. This not only caters for teachers working in the middle years but also allows primary school teachers to see where learners will be taken in their later mathematical learning.

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Larkin, K. (2016). Mathematics education and manipulatives: Which, when, how? *Australian Primary Mathematics Classroom, 21*(1), 12–17.

ABOUT THE AUTHORS

Robyn Jorgensen is an Emeritus Professor at the University of Canberra. Robyn has worked across the various sectors of schooling, from early years, primary, middle and secondary to vocational and post-compulsory schooling. Her research interests are centred on issues of equity and access, particularly for students from disadvantaged backgrounds—low socio-economic status, rural, remote and Indigenous students. She has received thirteen Australian Research Council grants to work in these areas. In 2009–10, she took up a position of CEO and Principal in a remote Aboriginal community. Her work is predominantly based in classrooms, where she works to better understand how teaching practices are implicated in the success (and failure) of learners. She has been Chair of the Queensland Studies Authority Curriculum Advisory Board for Mathematics; worked on the Queensland Ministerial Committee for Science, Technology, Engineering and Mathematics (STEM); and was the eminent Professor on the Australian Association of Mathematics Teachers' 'Count on Me' project, which seeks to improve mathematical learning for Indigenous Australians. Robyn has a distinguished publication record and has authored many books, journal articles and conference papers. She has been invited to be the keynote speaker at numerous state, national and international conferences. While currently living in Queensland, she has also lived in Victoria, New South Wales and the Northern Territory.

These experiences have helped to inform her breadth of knowledge of the diversity of Australian society and curriculum.

Shelley Dole is Professor and Head of School of Education at the University of the Sunshine Coast. Shelley is an educator and mathematics education researcher. Over the 25 years she has been in education, she has taught in primary, secondary and tertiary teaching institutions throughout Australia. Shelley has led two major Australian Research Council projects, focusing on numeracy across the curriculum and the development of proportional reasoning, with teachers and schools in Queensland and South Australia. She has been involved in several major research projects in Queensland, Tasmania, Victoria and South Australia, including middle-years literacy and numeracy; early-years literacy and numeracy and distance education; mental computation, number sense and invented algorithms; teaching and learning per cent in the middle school; basic facts in the early years; as well as professional development projects. In 2009, Shelley won a University of Queensland Award for Teaching Excellence, and in 2010 she was the recipient of an Australian Award for University Teaching. She is a senior fellow of the Higher Education Academy.

Kevin Larkin is a Senior Lecturer (Mathematics Education) at Griffith University, where he teaches two of the core primary undergraduate education mathematics courses at three campuses. Kevin is a mathematics educator with 37 years of educational experience. He taught in primary-school classrooms for fifteen years (both in Australia and internationally) and worked in various leadership positions for a further fourteen years. Since 2012, Kevin has worked at Griffith University. Every aspect of his teaching is informed by an overarching goal to inspire new generations of creative, motivational mathematics teachers. His teaching philosophy is based on a deep appreciation of the critical role of mutual and supportive working relationships between teachers and learners, and commitment to the provision of learning and teaching environments that are personally, socially and educationally rich. His teaching is complemented by a robust research and scholarship portfolio, which includes over 70 publications, two Office for Learning and Teaching grants, and three Griffith Grants for Learning and Teaching. Kevin has published widely in national and international publications in the areas of mathematics education, digital technologies, early-years STEM, higher education and activity theory. His particular area of expertise is the use of digital technologies in

mathematics education. He is a Senior Fellow of the Higher Education Academy, and is the Inaugural Chapter Chair for the Arts Education and Law Group Learning and Teaching Academy. Kevin's expertise in learning and teaching has been recognised, among other awards, with a National Teaching Citation in 2017 and recognition in 2016 as Griffith University's Teacher of the Year. Kevin is the 2018 University Teacher of the Year as well as the recipient of the 2018 Award for Teaching Excellence (Social and Behavioural Sciences).



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CHAPTER 1

THE CHANGING FACE OF SCHOOL MATHEMATICS

There is a need for new approaches to teaching mathematics. Sadly, mathematics is still rated by many students as one of their least favourite subjects at school, partly because mathematics teaching is too often typified by students working individually with their desks in rows facing the board, and the teacher demonstrating procedures from a textbook or an interactive whiteboard (IWB), with students completing worksheet exercises. With the vast amounts of research that have been generated since mathematics education became recognised as a discipline, there is now a strong research base to inform change in teaching school mathematics. Also, wide changes are occurring rapidly in societies, both nationally and internationally. Change impacts students, and there are new theories of how students learn in contemporary times. Research has also contributed to changing perceptions of mathematics as a discipline.

Mathematics education and society

The mathematics curriculum has not been created in a vacuum. Mathematics in schools, and the way it is taught, are the product of broader factors that extend beyond the classroom. Such factors include

employers, lobby groups, government policy, parents and professional organisations.

External authorities, such as education departments and other statutory bodies, may develop a curriculum or syllabus that provides guidelines teachers are expected to use to develop their work programs, and to undertake assessment and reporting of students' learning. The development of curriculum documents is influenced by demands from groups including parents, employers and governments. In the Australian context, the Australian federal government developed an agreement among the states and territories for the development and implementation of a national curriculum, which began in mathematics in 2012. Documents relating to mathematics in the Australian national curriculum can be accessed through the Australian Curriculum, Assessment and Reporting Authority (ACARA) website (www.australiancurriculum.edu.au/f-10-curriculum/mathematics/). Documents relating to the New Zealand National Curriculum can be found at the New Zealand Curriculum Online website (http://nzcurriculum.tki.org.nz/The-New-Zealand-Curriculum/Mathematics-and-statistics). When all variables are considered, the curriculum guidelines that appear in mathematics classrooms have been created through a highly negotiated (and often hotly contested) process. In addition to curricula, many jurisdictions around the world now utilise some form of high-stakes testing. Examples include, but are not limited to, the Trends in International Mathematics and Science Study (TIMSS), the Programme for International Student Assessment (PISA) and, in Australia, the National Assessment Program—Literacy and Numeracy (NAPLAN). The use of these external, high-stakes assessments affect how teachers teach, and students learn, mathematics.

In some cases, curriculum shaping is a reciprocal process, where the benefits of change are two-way. Consider the impacts of technology within society, and of numeracy expectations. In the current context, employers are demanding that students exit schools with high levels of numeracy (and literacy). As a result, there is a much heavier emphasis on numeracy in education. Similarly, schools recognise the value of technology as a learning tool that enables students to exit schools with a strong appreciation of how technology can be used to enhance mathematical work and thinking.

Teaching mathematics in modern society

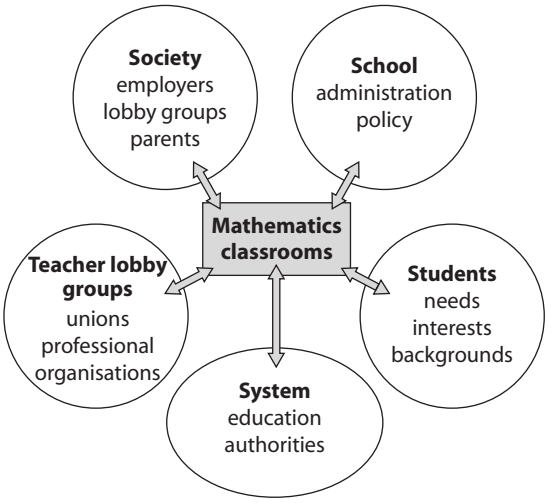
We use the term 'modern society' throughout this book to refer to the contemporary context of education. 'Modern' is more appropriate than terms such as 'Western' because it does not support a notion that Western views and approaches are more valid than those of Eastern or indigenous cultures. Instead, it suggests that the curriculum reflects a contemporary view of education that embraces new approaches to teaching, such as the use of digital resources (the internet, computers, tablet devices, and so on).

In the times in which our students live, technology, globalisation, the information age and very different patterns of family, leisure and work have brought changes to society, work, schools and life. We use the term 'modern society' to provoke thinking about the age in which we live and the quite different lives of contemporary young people, and to consider how these have changed since 'the old days'. Educational researchers have underscored that modern society is different. Many cultures—such as those that embrace Eastern philosophies, or indigenous cultures seeking to gain access to contemporary ways of thinking and learning—also live in the modern world. Curricula in schools must reflect the changes occurring in the wider society to ensure that schools adequately prepare students for the world beyond compulsory schooling.

Mathematics classrooms

Most young people are now growing up in technology-rich environments. They do not remember a time when you had to physically get up to change the television station (remote controls do that)—indeed, many are now watching television using streaming technologies. Cooking in pre-programmed microwaves happens at the touch of a button. Today's young people are generally technologically savvy.

One of the biggest growth areas in employment is self-employment, which means that many young people will be creating jobs for themselves in positions that will not even exist when they exit school. Our students are growing up immersed in an information-rich society—they no longer have to search through the relatively few books in a school or a local library, but can instead undertake searches on computers or handheld devices that may yield thousands or even millions of hits. The skills they



require in order to be able to search for and identify key information are very different from those they needed when only 'page-based' texts were available.

Students growing up in this technology-rich world have become used to multiple sources of information input—they are constantly bombarded with short bursts of infotainment, as well as brief snippets of information from television and other media. They are able to fragment and reconstruct images (such as maps) in ways unimaginable in former times (Lowrie 2003), as well as practise algebraic thinking through the study of curves made possible by dynamic modelling software (Padula 2014). Commonly used terms to describe contemporary students, such as 'cyberkids' or 'digital natives', although contested, recognise that their dispositions towards learning have been formed by the wider social conditions in which they have grown up. Traditional models of teaching and learning need to reflect, and sometimes to challenge, these changed circumstances.

The mathematics education of the students of modern society must be considered in light of this. Students need to develop mathematical ways of seeing and interpreting the world; they need to develop strong problem-solving skills; they need to be numerate; and, most importantly, they must have a disposition towards using mathematics to solve the problems they confront. School mathematics needs to adopt pedagogies that will cater for diversity within a classroom. The old models of seated individual work—found in what might be termed 'traditional mathematics teaching'—are possibly contributing to the problems that emerge as students progress through school. For considerable numbers of students in the years of upper primary and lower secondary school, the teaching that they encounter can lead to many negative feelings and misleading learnings about mathematics (Larkin & Jorgensen 2016).

Eastern philosophies that focus more on the wellbeing of learners are impacting the pedagogies used in Western classrooms. Notions of happiness and mindfulness, as per Buddhist philosophy, are gaining traction in teaching approaches. Yoga is finding a place in the classroom, too.

These approaches are assisting learners to focus and be 'in the moment', rather than be distracted by the numerous stimuli that abound in the contemporary world.

New models of teaching mathematics

The mathematics curriculum encountered by students prior to the 1960s focused on arithmetic and operations. Most of the mathematics education developed after that time in Western countries emerged post-Sputnik, when the race to the moon had become a race for intellectual superiority, with mathematics seen as the linchpin of success. The 'New Mathematics' contributed to a lock-step approach to teaching mathematics, with hierarchies in orders and sequences of teaching (Brown et al. 1998). The 1970s witnessed a boom in the ways in which mathematics curricula were organised; most were not research-based, but rather were influenced by arguments of logic and reason. A hierarchical approach to mathematics teaching ('skill, drill and kill', before application and problem-solving) was implemented in most Western classrooms, and for many teachers and systems, such an approach has become a way of life. Brown and colleagues (1998) argue that much of what was written in terms of mathematics curriculum reform had very little research base, thus raising questions about the validity of the curriculum itself.

In more recent times, there has been a growing awareness that such approaches are not resulting in positive learning outcomes. Indeed, as Clements (1989) argues, all that students learn from ten years of compulsory schooling (which, in most countries, may be extended to twelve to thirteen years) is that they cannot do mathematics! However, not all countries have bought into this approach to teaching. Research, while somewhat dated, emanating from these countries, particularly the Netherlands (see Anghileri 2001, 2006; Beishuizen 1999; Buys 2001; Treffers & Beishuizen 1999; van den Heuvel-Panhuizen 2001), has focused on developing new methods and approaches to teaching mathematics. This research is now being adopted in many countries, as it has been shown to enhance the understandings of numbers, particularly seeing numbers, in a very flexible way (Anghileri 2006; Revina & Leung 2018). The program, referred to as 'Realistic Mathematics' (van den Heuvel-Panhuizen & Drijvers 2014), has sought to develop deep number sense from real-world examples.

The Netherlands did not embrace the New Mathematics movement, instead focusing its efforts on how students think mathematically (Treffers 1991). There is now a substantive body of knowledge drawing on students' thinking that has not been constrained by New Mathematics. From this, curricula have been developed that draw on students' understandings, build on them, and move progressively towards abstract and formal mathematical processes. Dutch mathematics reformers call this 'progressive mathematisation'. Occurring in parallel with this work has been the development of constructivist theory and a general awareness that students actively construct meaning from their experiences. This latter work has had a powerful influence on mathematics education, with it increasingly being recognised that students' individual understandings are based on their lived experiences.

These twin movements have emerged at a time when it was being recognised that many of the old, 'traditional' methods of teaching mathematics were failing too many students. This made the moment ripe for identifying more valid methods of teaching mathematics, and many countries, states and provinces adopted new methods of mathematics teaching and learning. More recently, methods such as direct or explicit instruction have seen a resurgence in traditional behaviourist approaches to teaching, and bring their own set of challenges for the authentic learning of mathematics. The use of approaches such as direct instruction, which is founded on a strong behaviourist approach to teaching, is guided by the strong influence of conservative politics advocating a return to 'the basics', as well as the ongoing questioning of teacher quality. Teacher standards and curriculum/assessment requirements are now placing unprecedented demands on teachers. This makes the work of the mathematics teacher very complex and bound by many compliancy regulations.

Content and pedagogy

Contemporary approaches to teaching mathematics need to encourage two aspects: content and pedagogy.

Content is the intellectual integrity of the subject. It is where students learn, apply and appreciate mathematics, and where deep learning and deep knowledge are paramount to learning experiences. Importantly, students are able to make connections between the mathematics they learn and other curriculum areas, as well as with the world beyond school. It

is important for them to develop an appreciation of how mathematics is an informing discipline that has importance and relevance to many spheres of life.

Pedagogy relates to developing supportive environments where student diversity is recognised, and practices are developed that value and build upon the different backgrounds and knowledges that students bring to the mathematics classroom. Good pedagogy is about teachers developing inclusive practices to build and extend their students' knowledge and confidence in using and applying mathematics. Classrooms are places where students understand the expectations teachers have of them, and the work they are to undertake. In developing inclusive practices, intellectual integrity should also be preserved. Teachers need to value students and to believe that *all* students can learn mathematics.

Research on productive pedagogies (Education Queensland 2001; Mills & McGregor 2016) states that good pedagogy is about high intellectual engagement and helping students to see and make connections; it is learner-centred, with each individual's knowledge and culture valued, and students feeling supported in their learning. Many teachers bemoan issues of behaviour management in mathematics classes, but in a large longitudinal study on teaching (Education Queensland 2001), it was found that many of the elements of productive pedagogies were absent from the 2000-plus classrooms observed. Students were not engaged in deep learning about and through mathematics. Often, it was found that students undertook busywork during mathematics lessons (e.g. sticking butterflies on paper, rather than engaging in discussion about area) but did not engage in much, if any, deep mathematical learning. It must be asked whether students in mathematics classes engage in behaviour that is subsequently construed as 'misbehaviour' because they are bored or because the pedagogy is unsound.

Teachers can make a difference

Teachers and teaching can make a significant difference to students' learning outcomes in school mathematics. In a large study of effective teachers, Hattie (2014) reported that it was not the school that made the difference to students, but individual teachers. Teachers have a powerful influence over what and how students learn. Through the provision of an appropriate learning environment in which content and pedagogy

match the backgrounds, needs and interests of individual students, *all* students can learn mathematics.

The power of teachers' beliefs

In a study exploring the characteristics of effective teachers of numeracy, Askew and colleagues (1997) concluded that one of the most important influences on learning was the teacher's belief that all students could learn mathematics. Often, values and stereotypes influence how behaviours and actions are interpreted and implemented. Teachers who believe that some students, due to their backgrounds or behaviours, are unable to learn mathematics will ultimately create learning environments that construct the expected outcomes. This has been shown to be the case in many studies. For example, in a seminal study (Rosenthal & Jacobsen 1969), it was shown that when students were assigned scores randomly as they commenced study in a new class at the beginning of term, the teacher, believing that these were the students' academic scores, interacted with different students in particular ways. By the end of the teaching term, the students' results for that class correlated closely with the scores they had been assigned randomly at the term's beginning.

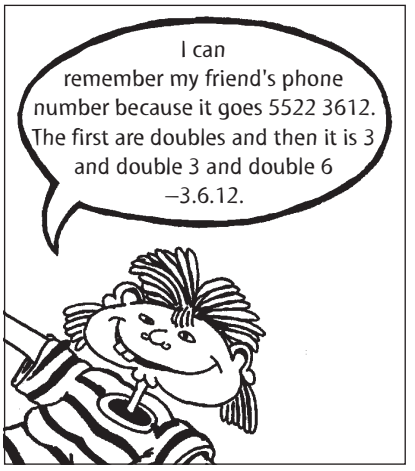
This study (and many subsequent ones) highlights how powerfully teachers' views of their students influence the ways in which they organise learning experiences. In an influential study undertaken in New Zealand, Bishop and Berryman (2006) investigated how to improve the educational achievement of Māori students, finding that if a teacher believes that all students can learn mathematics, then learning environments are likely to reinforce this belief. The important work undertaken by Sarra (2012) and others (e.g. Buxton 2017) in relation to Aboriginal and Torres Strait Islander learners has focused on teachers having high expectations of learners. Similarly, if teachers believe that the best way to learn mathematics is by making authentic connections with real-life examples, or by incorporating digital technologies to support students' mathematical understanding, then their teaching and learning environment will reflect these beliefs. The classroom is likely to be peppered with equipment and displays demonstrating the links between mathematics and the world beyond schools. In good teaching, teachers must be mindful of the influence of their beliefs about teaching and learning mathematics, and of the influence these have on their students' learning outcomes.

What is mathematics?

When a member of the general community is asked the question, 'What is mathematics?', they invariably focus on number and operations. However, the modern mathematics curriculum is far more complex than arithmetic alone. School mathematics has changed over time; during different periods, different mathematics has been taught. A century ago, mathematics in upper-primary classes entailed computation of tasks involving a large amount of numbers, long division, square roots of non-square numbers, and so on. This form of curriculum remained in place until the early 1960s, when the implementation of the New Mathematics represented a considerable shift in mathematics curriculum in most Western countries. New topics were included, and new forms of thinking mathematically (e.g. set theory) became part of the new syllabi.

Since the 1970s, other reforms have influenced the curriculum, including problem-solving in which students were expected to be more creative in their thinking. In the 1980s and, increasingly, since then, technology has played a role in the curriculum. While the software program LOGO® had a strong influence in the early years of technology-aided learning, it has now been replaced by other technological aids, including new software programs, spreadsheets and graphic calculators, as well as, more recently, a multitude of applications (apps) available on tablet devices. Students are expected to be far more creative in their thinking, and to deal with much more knowledge and complexity than in the pre-1960s era.

Mathematics is the study of patterns and relationships



One of the defining characteristics of mathematics is awareness of recurring ideas and relationships between and among mathematical ideas. The knowledge learned in one area of mathematics links to other areas—for example, the number facts do not have to be learned as discrete facts; rather, the student who can see that when you add 3 and 5 there is a total of 8 can also recognise that if you have 8 items and 3 are lost, then there will be 5 remaining. The place value system can

be seen as a patterning of 10, so that allowing young students to play on a calculator or a tablet device with the constant function set for the addition of 10 means they can see the pattern of adding tens occurring in the tens column. Seeing these types of patterns and relationships is a key factor in how students come to learn and appreciate mathematics.

Mathematics is a way of thinking, seeing and organising the world

Students who learn mathematics as a dynamic discipline through which much of the world can be interpreted are able to make sense of a wide range of experiences. They are able to organise and analyse events in systematic ways. Sometimes this might be numerically, but it could equally be spatially or through logical reasoning. Viewing the world through a mathematical lens can make everyday tasks, such as remembering phone numbers, easier, because a pattern can be seen and people are thus able to solve problems more efficiently and effectively. Mathematics moves beyond the memorisation of a vast body of facts and procedures in a rote manner to become a more systematic and insightful process. This ability to think, see and organise the world using mathematics is particularly important in this age of 'big data', where we are swamped with information.

Mathematics is a language

In past times, mathematics and language were seen as two disparate disciplines. Today, it is seen that, in order to learn, appreciate and understand mathematics, students need to learn the language of mathematics—complete with its unique words, grammar and symbols. Knowledge of the language of mathematics allows complex ideas to be communicated easily. Language involves communication, and mathematics is a very particular and precise language that is communicated in particular ways. It involves the contraction of lengthy tasks into concise 'sentences'. For example, a shopping expedition during which three items were purchased in one outlet and five in another becomes $3 + 5 = 8$. Abstract symbols, such as 3 and 5, represent specific concepts.

Mathematics is a tool

People use mathematics to solve problems every day. The more competent a person is with mathematics, the more efficiently they can solve problems

and, in many cases, the more they can thrive in the world beyond school. Deciding which goods to purchase, checking bank balances, being able to budget, and deciding which phone plan or mortgage to take out or how much concrete to purchase for a patio all require the use of mathematics. Mathematics enables people to make sound decisions and judgements, and to solve problems.

Mathematics is a form of art

For many students, mathematics is a grind that has to be done under duress. In contrast, for those who have been fortunate enough to engage with mathematics, its internal beauty is awesome to behold—it has the consistency of a fine artwork. The logical coherence of Fibonacci's numbers, for example, is almost magical. (The Fibonacci sequence is generated by adding the two previous numbers in the sequence to get the next number: 1, 1, 2, 3, 5, 8, 13 . . .). Likewise, the tessellations incorporated into the artistic work of Escher remind us of the beauty of mathematics. This aspect of mathematics is perhaps the most elusive for many people, since they equate mathematics with the school process of 'doing mathematics' rather than viewing it as something to be appreciated.

Mathematics is power

Mathematics has been behind most inventions—good and bad—in modern history. It has enabled us to walk on the moon, but also to invent nuclear weapons with their potential for mass destruction. Mathematics also gives you the power to take control in your life by ensuring you are given the right change in money transactions, questioning information presented in mathematical terms, and so on. Those students who succeed in mathematics are more likely to live successful lives than those who do not. Mathematics is the social filter that facilitates the access of some students to professions of high status, wealth and power while excluding others. Developing nations actively seek access to mathematics for their youth because they know that such knowledge will benefit them in the future. Many Western countries are recognising the problems inherent in the decreasing number of students undertaking formal studies in mathematics, which is the foundation for many other forms of powerful knowledge. STEM-related disciplines such as computing, science, engineering and technology, for example, all rely heavily on a solid understanding of mathematics.

Where does mathematics come from?

While there are a number of theories regarding the generation of mathematics, two key but opposing views permeate the discipline. While there are other views—dealt with in considerable detail by Ernest (1991)—it is worth considering the two main views and how they impact both the ways in which mathematics curricula are constructed and how the discipline is taught. Just as theories of how humans developed create considerable debate, so too do theories about the origin of mathematics. There are both Platonists and Constructionists in mathematics. Some of their opposing notions are listed in the table below.

Opposing mathematical notions in Platonist and Constructionist theory

Platonist	Constructionist
Mathematics is pre-existing	Mathematics is a construction of knowledge
The role of the mathematician is to 'discover' facts	Mathematicians create new knowledge in order to explain or expand ideas
Mathematics is about truths and objectivity	Mathematics is about creativity and subjectivity
Teaching involves learning of facts	Teaching involves creating learning environments for students to create their own mathematical understandings
Assessment is objective and tests for factual knowledge; e.g. $\frac{3}{10} + \frac{4}{10} = \frac{7}{10}$ is assessable and true—it is either right or wrong	Assessment is subjective; e.g. $\frac{3}{10} + \frac{4}{10}$ should be $\frac{7}{10}$ but if a student responds with $\frac{7}{20}$, question why. It may be that the student sees the question as being about test results from two exams and not the addition of two fractions

The mathematics curriculum

Teachers developing school-based programs or shared units of work can attest to the debates that develop in seeking to form a coherent learning scheme. The mathematics curriculum is a socially negotiated and constructed document; as such, it represents particular viewpoints of various interest groups. The curriculum that is presented by a particular education authority or through a commercial venture is the product of a group of people working towards the development of an agreed document.

The meaning of the term 'curriculum' is highly contested, ranging from the collection of knowledge that will be implemented formally in a classroom through to the planned and unplanned learning of students coming to school. The more widely agreed notion of curriculum is that of knowledge being implemented so that it resonates with the notion of syllabus. This is then supported through teaching and assessment practices.

Curriculum design

When planning mathematics curricula, departments of education and other peak bodies tend to organise them around a number of key features. In many cases, these are seen primarily as the content areas, but mathematics encompasses far more than this. Consequently, these documents often include pedagogical approaches to teaching and assessment; principles of equity and diversity within the mathematics classroom and how best to cater for this aspect of mathematics education; and contemporary shifts in practice that reflect the dynamic and changing nature of not only the content but also ways of teaching and assessing mathematics. Often, the role of various technologies in mathematics is also included.

Curriculum organisation

Contemporary terminology for organising mathematics sees the curriculum broken down into content areas and levels of difficulty. These general principles for organising curricula have filtered through to most contemporary international curriculum documents, and in Australia since the early 1990s, topics within the mathematics curriculum have generally been organised under five main headings, identified as *strands*. Before 2012, the five strands of the curriculum were Number, Space, Measurement, Chance and Data, and Algebra. The curriculum frequently includes reference to 'working mathematically' to capture the processes and attitudes that the mathematics curriculum seeks to promote. In Australia, a new national mathematics curriculum was launched in 2012. It is organised according to six strands, which have been paired: Number and Algebra, Measurement and Geometry, and Statistics and Probability. The overarching idea of 'working mathematically' has been replaced by four proficiencies: Understanding, Fluency, Reasoning and Problem-solving. Similarly, in New Zealand the curriculum is organised according to three strands: Number and Algebra, Geometry and Measurement, and Statistics (including probability). Increasingly, the role of digital

technologies in mathematics is acknowledged in curriculum documents. For example, in the Australian Curriculum: Mathematics, the use of digital technologies appears in a number of content descriptors across the primary years.

A general overview of the sorts of topics assigned to these strands is shown in the following table.

NUMBER
- Pre-number work around patterning, one-to-one correspondence
- Number work—cardinal numbers, nominal numbers, ordinal numbers, counting, place value
- Whole-number operations (addition, subtraction, multiplication and division)
- Rational numbers—fractions, common, decimals, percentages, mixed
- Estimation, mental computations
- Number sense
- Ratio and proportion

ALGEBRA
- Pre-algebra patterning
- Equality
- Algebra
- Functions

STATISTICS
- Data representation and interpretation

PROBABILITY
- Chance
- Statistics and data—collect, organise, display

PROFICIENCIES
- Understanding, Fluency, Reasoning and Problem-solving
- Attitudes and values
- Mathematical inquiry
- Choosing and using mathematics (modelling, applications)
- Communication, representation and language
- Problem-solving
- Connections with other strands, curriculum areas, beyond school

MEASUREMENT
- Length
- Area
- Volume and capacity
- Mass
- Time
- Money
- Temperature

GEOMETRY
- Location and arrangement—directions, grid work, coordinates (topological)
- Shape—2D shapes, 3D solids, structures, properties, tessellations
- Transformation, symmetry—changes in position, location
- Geometric reasoning

A model for organising curriculum content areas

Number

This refers to the study of pure number. It involves the early study of number and patterning. In order to develop good number sense, understanding how patterns are developed is an important part of mathematical thinking. The early years of schooling place considerable emphasis on developing a good sense of number. Formal study of number and operations occupies a considerable amount of space in the timetable. Teaching

of whole numbers and operations on whole numbers generally precedes work on part numbers. Given the importance of number, it is likely that a large proportion of a teacher's time will involve the teaching of number.

Algebra

In some countries, algebra does not appear in the primary-school years, and is only introduced when students move into secondary school. In part, this could stem from the Piagetian ideology that primary-school students are not able to think abstractly; since algebra is highly abstract, it is seen as being beyond the scope of younger students. However, algebra is about patterning, and being able to identify and describe patterns is integral to the mathematics curriculum and lays the foundation for algebraic thinking (Maj-Tatsis & Tatsis 2018).

Measurement

This strand seeks to measure objects. There is potential for considerable slippage in this strand, since the concept of measurement relates to other strands as well. For example, angle could rest as easily in the measurement strand as in the geometry strand. In some documents, money—a measurement of value—is placed in the number strand because it links strongly with students' experiences with number, and notions of decimal fractions are applied easily to number. Similarly, chance and data can be measured, so in some documents they are located in this strand. These differences show the arbitrariness of curriculum construction.

Geometry

This strand, the study of how space is filled and described, typically includes shape, geometric reasoning, location and arrangement. This element of the curriculum tends to be slightly more fragmented than the other strands, since the sub-strands are not so connected, and hence are less open to a coherent framework for development. A considerable body of research documents the importance of geometry in later school achievement in mathematics and beyond.

Statistics

This is the strand that works from data to make inferences. It requires students to identify appropriate ways to collect, analyse, interpret and make inferences from data. An increasing sophistication is expected of

learners as they progress through the strand, so that they can analyse data and critically appraise statistical information provided through a range of media. A branch of mathematics that is commonly associated with this strand is one referred to as 'Critical Mathematics'. This branch of mathematics seeks to identify the politics behind the use of statistical data.

Probability

Probability is often based on data collected and used to predict outcomes. In order to make effective predictions, data are usually collected or inferred mathematically, and presented in a format that makes sense of the data so that informed decisions can be made about the predictability of an event occurring. There are two main approaches to probability—experimental and theoretical—which students apply to situations and assess the likelihood of events occurring. Both approaches are used in the teaching of probability. Since data are collected and the chance of an event is predicted, some curriculum documents have statistics and probability as topics in the measurement strand.

Working mathematically (proficiencies)

Contemporary thinking sees mathematics as more than a collection of knowledge and skills. It is instead viewed as a way of thinking and working—a new dimension that is being included in most modern curriculum documents. Debate on the inclusion of mathematics as a separate curriculum strand is something that teachers need to address. The argument for mathematics to be integrated is founded on the belief that it is very difficult to undertake much mathematics without working mathematically; thus, having it as a separate strand would result in considerable doubling up of content. For example, an operation such as multiplication is often raised in a problem-solving context; while the two items are covered simultaneously, the teacher may focus on only one outcome. Not seeing the operation holistically results in considerably more work.

Diversity in curriculum organisation

While there is considerable diversity in the organisation of mathematics curricula in various states within individual countries, and across different countries, there is enough synergy among them to talk in general terms. Typically, curriculum documents are broken down into a number of organising concepts. The content areas discussed above are not meant

to be comprehensive; rather, they provide some guidelines about what could typically be expected in a strand. Rather than being perceived as a static organiser, such an outline should be seen as a set of organisers that interrelate with each other. For example, multiplication links strongly with area and volume; negative integers link with temperature; and money is a measure, but also links with decimal understanding (the Number strand). Because documents are working and evolve over time, different manifestations of them can reasonably be expected, since writers and developers work under very different constraints. As curriculum documents are developed, demands are placed on writers to conform to other expectations—such as those of teachers who will be using the documents, policymakers, systems administrators, parents, employers and other lobby groups. The curriculum should be seen as a representation of particular values and interests within the community from which it emerges.

As indicated earlier, this process occurred in Australia from 2012 via the development of a new organising framework for the mathematics curriculum, reducing the five strands to three paired strands (Number and Algebra, Measurement and Geometry, and Statistics and Probability). The rationale for this organisation was that it would enable greater connectivity between topics within strands. The Australian mathematics curriculum has integrated proficiency strands that, in a sense, envelop but extend a strand of working mathematically. There are four proficiency strands: Understanding, Fluency, Problem-solving and Reasoning. The *Understanding* strand emphasises the conceptual aspects of the content, and *Fluency* emphasises associated skills. *Problem-solving* is about strategies, communicating mathematically and interpretation of problems, while *Reasoning* incorporates processes of justifying, explaining, analysing, and so on. Curriculum organisation is thus changing and shifting, and reflects the changing society in which we live and the social construction of mathematics curriculum documents.

Technology for thinking mathematically

The changes in the wider society towards a much more technology-oriented lifestyle have resulted in significant changes in the ways in which mathematics is developed. There is a substantive body of knowledge that demonstrates how computers and calculators encourage much deeper

levels of mathematical thinking than are possible with pencil-and-paper work (e.g. Gowland 1998). For example, in their work with calculators, Groves (1995) has shown that students can gain a greater sense of number in the early years through guided play with calculators. Whereas most curriculum documents demarcate the first year for number study of 1 to 10 (or maybe 1 to 20), the following year for numbers to 100, and the subsequent year for numbers to 1000, Groves has shown that students in their first year of school can gain a strong grasp of numbers up to 1000. In part, this is due to a reduction of other demands—such as writing numerals—and a greater emphasis on thinking mathematically. Other advocates of digital tools have shown that their effects have been significant, both as a computational aid and as a learning tool (Bennison & Goos 2010). However, opponents of calculators (and, more recently, tablets and computers) have argued that they reduce students' capacity to complete even simple tasks. Longstanding research demonstrating the capacity of digital tools has been resisted by educators and policymakers who, unfortunately, see the digital tools as a replacement for mathematical thinking rather than as a tool to assist and extend learning.

The role of graphic calculators and graphics software in secondary-school contexts has also been debated on much the same grounds. Studies have shown that graphic calculators have enormous benefits in improving deep understanding of functions and algebra (Doerr & Zangor 2000). This is due to the calculators' capacity to show the graphs, as well as how the gradients and intercepts change with changes to the function. This dynamic process allows students to see the effects of changes quickly; old methods focused on physically plotting the graphs, which meant that few graphs could be undertaken, thereby restricting the dynamic visualisation process.

The computer and its applications have had similar effects on mathematical thinking, fostering deeper understandings of mathematical ideas. For example, the graphing function allows students to construct bar graphs or pie graphs painlessly. Constructing pie graphs by hand is very tedious. Spreadsheet software allows students to explore which graphs are best for which purposes in ways not possible (or at least restricted) when constructing graphs through pencil-and-paper methods. Similarly, using spreadsheet functionality requires students to develop simple instructions for calculations—such as 'add cell B1 to cell B2'. This process requires some algebraic thinking, and provides a strong rationale for using algebra.

See Calder (2010) for further discussion on the usefulness of spreadsheets in supporting mathematical thinking.

More recently, digital manipulatives have become increasingly available for use in mathematics classrooms (Bray and Tangney 2016). Two useful sites for teachers to access manipulatives are Illuminations (https://illuminations.nctm.org/) and the National Library of Virtual Manipulatives (http://nlvm.usu.edu/en/nav/vlibrary.html). Many of these manipulatives are now also available for use on tablet devices. The advent of more sophisticated digital tools makes it possible to pose deeper questions about mathematics—questions about patterns and place value, as well as higher-order questions such as 'Why?', 'What if?', 'What happens when?' and 'Why is this happening?' The focus shifts from the process of construction to understanding the reasons, purposes and rationales for undertaking particular work. For instance, research by Larkin, Kortenkamp, Ladel and Etzold (2018) investigated the potential of young children, supported by an application (https://apps.apple.com/au/app/place-value-chart/id568750442), to understand place value at a deeper mathematical level than that currently proposed in many curricula.

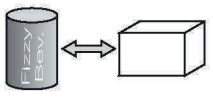
Young people in modern society have grown up in a technology-rich environment. Their familiarity with technology makes it an amenable tool for considerable innovations in mathematics education. Software such as spreadsheets can be open-ended, offering considerable potential for use in many ways to support various aspects of mathematics. Other packages are closed and restricted to particular topic areas. Examples include specific software packages for learning multiplication facts or other operations (e.g. Maths Blaster®). Such packages can simply be upmarket worksheets, and hence encourage rote learning rather than mathematical thinking. Packages such as LOGO®, Geometers Sketchpad®, GeoGebra or Calibri may be oriented towards particular topic areas—in many cases, geometry—but still have an open-endedness to them, allowing students to explore other aspects of mathematics, particularly working mathematically. When selecting software, teachers should be cognisant of what they contain and mindful of their shortcomings. Scootle (www.scootle.edu.au/ec/p/home), an online database of digital resources, provides a reliable source of activities for teachers that are mapped against curriculum topics. Overall, the research suggests that digital technologies, including calculators, are powerful tools for supporting, enhancing and extending students' mathematical understandings when used appropriately in the classroom.

Integrating curriculum

The mathematics curriculum gains its benefit and purposefulness as a discipline that informs other disciplines. For example, within the social studies curriculum, where students study environments and demographics, there is frequent referral to graphs and charts; without the contributing knowledge gained in mathematics, students would be restricted in their capacity to interpret this information. One of the major concerns of mathematics education is the fragmentation of knowledge, which leads to teachers and students not being provided with learning experiences that allow them to make connections between concepts. For students, this can result in their perceiving mathematics as a large discipline comprising a lot of disconnected facts. For teachers, it can result in the planning of many activities so that considerably more time is spent achieving outcomes. Thus, it is important to see the connections between the various strands of the mathematics curriculum, the connections to other curriculum areas and the connections to the world beyond school.

Relationships with other areas of mathematics

Each of the various strands of a mathematics curriculum should not be seen in isolation from the other strands or sub-strands; their links to other strands or sub-strands should be considered. For example, it is difficult to teach measurement concepts and skills unless students understand other number concepts (such as 1000), understand multiplication processes and have some sense of number. Similarly, the relationships between the sub-strands of volume and capacity can be explored (many students are not aware of the relationship between mL and cm, for example). One of the disadvantages of breaking knowledge into compartments, or fragments, is that such connections are at risk of not being noted.



The relationship between liquid and solid measures

Integrating mathematics with other curriculum areas

Many teachers practise using an integrated curriculum model, and subject integration is a signature practice of the middle years of schooling (or

MYS) pedagogy. This means that teachers link a number of curriculum areas together. There are three reasons for working in this way:

1. *It works in a pragmatic sense.* With the increasing amount of content to be covered in primary schools, and the increasing demands on teachers' time, teachers have fewer resources available to cover essential learnings. Integration across curriculum areas allows them to ensure that they make the best use of their time. For example, if a teacher is covering report-writing in English, it is possible to link this to science and expect students to write a report for an experiment that has been conducted. Such an experiment may have involved considerable mathematics in measuring results, so links can be made to the measurement strand. Such cross-curriculum links allow teachers to assess and access students' understanding in three areas, rather than one.
2. *Mathematics cannot exist on its own.* One of the criticisms of mathematics teaching can be that it is divorced from other areas of learning. Making links to other curriculum areas allows students to see how the skills they are learning in mathematics are relevant to other areas of study. For example, students may be monitoring the water quality in a local creek. This is a part of the environmental science unit. The data that have been collected each day or week can be recorded as tables or graphs and then interpreted to determine the quality of the water. Using their mathematics skills in collecting and recording data, students are able to see their graphing skills as having a real purpose. In another example, students plotting points on a globe require a working knowledge of spherical geometry; thus, the application of mathematics is central to being able to identify locations through coordinates. This is a critical aspect of good teaching of mathematics—students being able to see why they need to learn mathematical skills and knowledge.
3. *Mathematics connects to the world beyond school.* Studies on the transfer of mathematics from the school to the workplace have shown that very little is, in fact, transferred between the two contexts. Workers often fail to use the knowledge that they learned in school mathematics to help them in their work and home environments, and this can restrict the effectiveness and quality of their lives. For example, calculating bank

balances or working out the best buys in the supermarket requires the knowledge and skills learned in school; these are experiences in which students can make important links to the world beyond the mathematics classroom. Being able to use and apply mathematics in daily activities empowers people by enabling them to make informed decisions.

Critical Mathematics

There is an approach in mathematics that specifically considers the political implications of mathematics education. Mathematics, especially when viewed from an objectivist or Platonic view, is often seen as something apolitical. That is, mathematics is an objective set of facts. However, from a Critical Mathematics standpoint, mathematics can be very political. When teaching fractions, for example, it is seen to be mathematically appropriate that sharing is an equal division of quantity. However, in some cultures sharing is based on the status of the people, and older people may get a larger share than younger people. The teaching of per cent is often done within the context of the marking up (or down) of prices. The inflation of prices to reflect profit and loss masks the capitalist ideology that profit is a normal part of society. Critical Mathematics questions the notion of normalising profit, and asks questions of who wins and who loses in these contexts. A similar example is students undertaking a large investigation focused on litter or waste. Rather than leaving the investigation as a set of tables or charts to show the findings, a further action-oriented step is undertaken to address the issues identified—such as writing to a member of parliament or taking other direct action in relation to the issue. Many examples of Critical Mathematics can be found in web-based searches.

Review questions

1.1 What are your experiences of school mathematics? How would you categorise your school mathematics experiences: Platonist or Constructionist? Give examples of experiences to support your position.

1.2 How is the mathematics curriculum organised? Write a summary of the key emphases for each strand and/or sub-strand within the mathematics curriculum in your educational context.

1.3 What is the place of written algorithms, such as long division, in your school curriculum documents? What do the current syllabus documents say about the teaching of written algorithms? What is your view on this?

1.4 What is the place of technology in the school mathematics curriculum? How has technology impacted the way you learn mathematics? What do new technologies mean for school mathematics?

Further reading

Historical curriculum documents are a good way to learn more about curriculum organisation, content and emphases. You should also look at the following:

- Curriculum documents of your locality. These may be local, state/territory or national documents. Frequently, state/territory authorities will publish curriculum or syllabus documents that interpret national documents, to which teachers are expected to adhere as they plan their teaching. All teachers need to be familiar with the materials relevant to their contexts.
- Other support materials. These can be from commercial publishers and/or can be departmental resource materials developed by local authorities to support teachers in schools.
- Materials from other regions (nationally and internationally) so that a broader sense of curriculum can be developed.
- Australian Education Council (1991). *A national statement on mathematics for Australian Schools*. Melbourne: Curriculum Corporation.

In addition to curriculum and support documents, these readings will also be useful in supporting your understanding of mathematics:

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CHAPTER 2

THEORIES OF LEARNING MATHEMATICS

Why study theories of learning mathematics?

To be able to plan how to teach mathematics effectively, there needs to be some understanding of how students learn mathematics. A major United Kingdom review of effective teachers of numeracy (Askew et al. 1997) has shown that theory is one of the key factors in developing high-quality practices and outcomes in numeracy learning. The authors argue that 'teachers' beliefs and understandings of the mathematics and pedagogical purposes behind particular classroom practices seem to be more important than the forms of practices themselves' (p. 3). That is, it is not so much the ways in which teachers use particular practices in the classroom (e.g. whole-group work, mental maths, direct teaching or other methods) that are critical, but rather the beliefs they hold towards their teaching approaches. The role of theory in underpinning practice is an essential element of quality teaching in mathematics. The ways in which teachers interpret students' mathematical responses and behaviours are shaped by their theoretical assumptions of learning (Jorgensen & Larkin, 2017). The same responses can be interpreted quite differently

by teachers depending on how they see the learners and learning. One of the clearest examples of the impact of theory is the attempt to understand why some students are more at risk of failing mathematics than others (see Chapter 4). If a teacher subscribes to a cognitivist view of learning, they may ascribe failing to some innate (and deficient) thinking process. In contrast, if a teacher subscribes to a more social theory of learning, they may recognise that learners come to school mathematics with very different backgrounds and languages, and that these influence what is valued in mathematics; hence, there is a disjuncture between school and home knowledge systems, and this is the potential cause of failing. Coming to a better understanding of the roles of theory in shaping thinking is central to better understandings of practice. In many cases, theory may not be stated explicitly or known, and will remain at a level of intuitive understandings. Articulating these understandings, in the form of theory, is the work of good teaching.

While a number of theories are observable in mathematics education, it is possible that various theories may be applied at different times, or in tandem with one another. For example, in the teaching of a skill or process where the ultimate aim is to achieve a high degree of automaticity (fluency) (e.g. the multiplication facts), behaviourist approaches might be applied during practice activities. In contrast, prior to practising number facts, where there is a need for deep learning of number (e.g. developing a sense of number), a more prudent approach to teaching would be to adopt a constructivist approach, where the need is for deep, interconnected understandings. (This is the approach outlined in Chapter 9.)

By having an idea of how students learn, teachers are better able to plan and anticipate in particular ways, and to create learning environments to facilitate better learning. Three significant classes of theory have had a strong influence on our understandings of how students come to learn and understand mathematics:

1. cognitive theories that focus on students' thinking
2. sociocultural theories that seek to understand cognition within a social context
3. social (or socially critical) theories.

This chapter provides an overview of key ideas associated with these theories. Theories associated with learning in modern society are also discussed.